



Society of Petroleum Engineers

SPE-229854-MS

Analytical Approaches to Subsea Pipeline Stress Analysis: Energy Methods as an Alternative to Numerical Methods

Mohamed Hermas, MCS Group, Cairo, EGY

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

The process of subsea pipeline stress analysis is inherently complex, often necessitating the use of finite element method (FEM) software, particularly in "large displacement" conditions. However, for scenarios in which the deflection angle remains below 10 degrees, this study explores fundamental analytical methods as an alternative to FEM.

Through the application of energy methods, the research addresses the frequently encountered scenario of single-point lifting above the seabed. Upon determining the pipeline's profile, tension, and bending moments, subsequent analysis employs the mechanics of materials to compute bending, tensile, and equivalent stresses. The equivalent stress is then evaluated and compared to the allowable material yield strength, enabling engineers to make design decisions aligned with established codes and standards.

This work integrates case studies to validate the proposed methods by comparing their outcomes with results obtained from commercial FEM software. Additionally, the study introduces a prototype software tool designed to deliver instantaneous results upon input modification. Unlike FEM, the proposed program's underlying equations demand significantly less computational effort, offering an efficient alternative for commercial software.

The accuracy of the proposed method is demonstrated to be approximately 99%, which is highly safe, especially when the total stress remains well below allowable stress. However, accuracy diminishes notably when the rotation angle exceeds 10 degrees and becomes critically unreliable at angles greater than 15 degrees, necessitating caution in such cases.

In conclusion, this study underscores the viability of basic analytical methods for pipeline stress analysis, providing a practical and cost-effective alternative to FEM when the pipeline rotation angle is low. By advocating for context-appropriate methodologies, the research aims to enhance the selection of analytical tools, fostering efficiency without compromising accuracy in engineering design.

Introduction

Submarine pipeline stress analysis is based on modelling the pipeline as a slender beam, fixed at both ends, and primarily subjected to large displacements. Over the past four decades, several commercial software programs have been developed, specializing in pipeline stress analysis. These programs have evolved to

encompass a broader range of scenarios, including free spans analysis and on-bottom roughness assessment. The capabilities of these software programs have advanced to the extent that they can address ad-hoc situations, such as single-point lifting, which is required for connecting a pipeline to a tie-in spool piece.

However, this paper demonstrates that in many instances, stress analysis can be effectively conducted without the reliance on commercial software. This approach optimizes the time and effort required to perform engineering tasks and positively impacts the overall performance of organizations.

Deflection angle

The deflection angle (or the rotation angle in continuum mechanics terms) is a significant measure of the degree of nonlinearity in structural mechanics. In studying the fundamentals of structural mechanics, it is typically assumed that the rotation angle is infinitesimally small, ensuring that the location of the external force remains unchanged as the body's shape changes. Consequently, the force-displacement relationship remains linear, as the geometry of the structure is nearly the same. This situation is addressed under the infinitesimal theory of beams, which serves as the foundational framework for structural and mechanical engineering students before they encounter more complex problems. Accordingly, as the deflection angle increases, the assumption of linearity becomes increasingly questionable.

The pertinent question arises: what constitutes an excessive deflection angle? In other words, what are the boundaries of the linearity in terms of the deflection angle?

As will be indicated in the subsequent section, in certain subsea construction scenarios, a structure deflected by 5 degrees can still be considered as a linear problem, i.e., it can be addressed under the infinitesimal beam theory. Conversely, in other cases, a deflection angle exceeding 1 degree may be categorized as a ‘moderately’ nonlinear problem. However, beyond a specific limit, the problem must be regarded as ‘fully’ nonlinear. Generally, structural problems are classified into three categories (in terms of nonlinearity) in some textbooks (Byskov, 2013):

- i. Small deflections: In this category, the load is linearly related to the displacement, and the equations of equilibrium are derived and solved using straightforward methods.
- ii. Moderately large deflections: The load-displacement relationship is nonlinear; however, the differential equation of equilibrium is simplified as the rotation angle is still small enough to cancel quadratic terms
- iii. Large deflections: In this category, no closed-form solutions are available, necessitating the use of numerical methods, such as the finite element method (FEM).

Most of the stress analysis scenarios in subsea pipelines fall within the first two categories. The pipelay analysis is likely the only case where FEM is indispensable. Therefore, the boundaries of nonlinearities are primarily determined by engineering experience specific to each field (Byskov, 2013).

Single-point Lifting

Background

In certain circumstances, it becomes necessary to elevate the pipeline end slightly, potentially for purposes such as repositioning or flange alignment. In such scenarios, concerns may be regarding the overstressing or buckling of the pipeline. Generally, based on engineering principles, there is no significant issue with lifting the pipeline one or two meters off the seabed, particularly if the operation is conducted gradually. Nevertheless, it is considered prudent engineering practice to perform a lifting analysis, typically utilizing software such as OFFPIPE, Orcaflex, or other commercial programs, prior to proceeding.

This paper will focus on the single-point lift scenario to demonstrate the primary concept, as it serves as the most illustrative example, highlighting the impact of the rotation angle on the choice of analysis method. The figure below illustrates how the pipeline is lifted to install a spool piece.

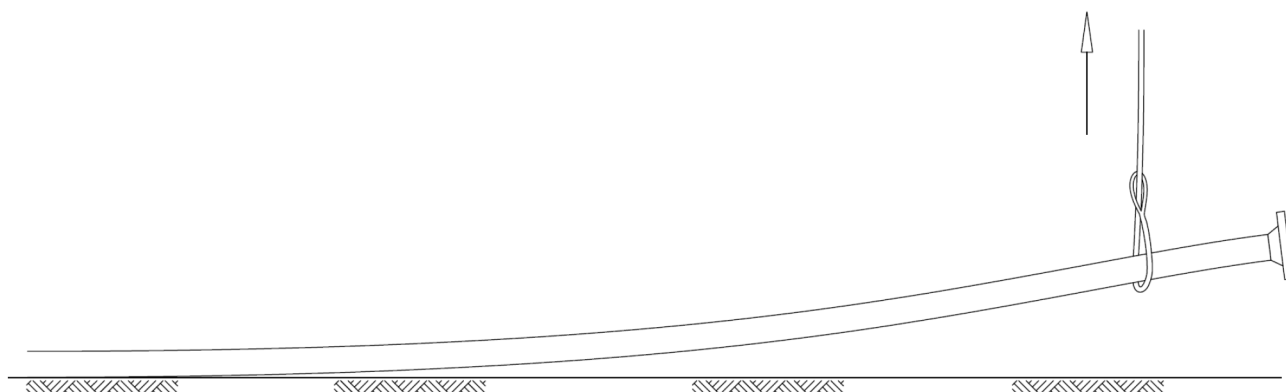


Figure 1—Single-point lifting schematic

With the widespread adoption of FEM-based commercial software (such as Orcaflex and OFFPIPE), all the engineering requirements are addressed, including the simplest scenarios (like single-point lifting), by considering full nonlinearity. As shown in the figure above, the rotation angle is approximately 5 degrees, indicating that the stress analysis can be conducted using analytical methods (rather than numerical methods), thereby conserving a significant amount of computational effort and cost.

However, this does not imply that the (simpler) analytical methods should be performed manually, as this would be more time-consuming than using sophisticated commercial software. Consequently, it is advisable to develop a spreadsheet (or a small application, as will be demonstrated later) that executes the analysis using the equations of simple analytical methods.

Touchdown point (TDP)

As illustrated in Fig. 2, the downward distributed loading (q) represents the pipeline weight per meter, while the concentrated load (P) signifies the lifting point, which could be connected to a crane or an air bag. The diagram further delineates the following lengths:

L_f : the distance between the lifting point and the TDP, which is an unknown variable.

$dtip$: the distance between the pipeline tip and the lifting point, judiciously selected by engineers (input data).

L : total length from the pipeline tip to the TDP, which is dependent on the two variables above.

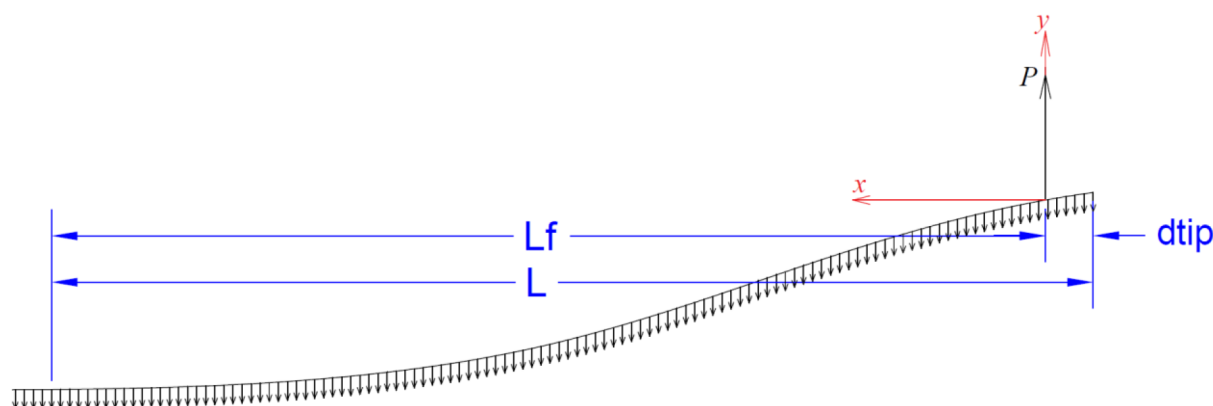


Figure 2—loading diagram

To calculate the touchdown point (TDP) distance, we employ a free-body diagram (FBD) of the right segment of the pipeline as depicted in Fig. 3.

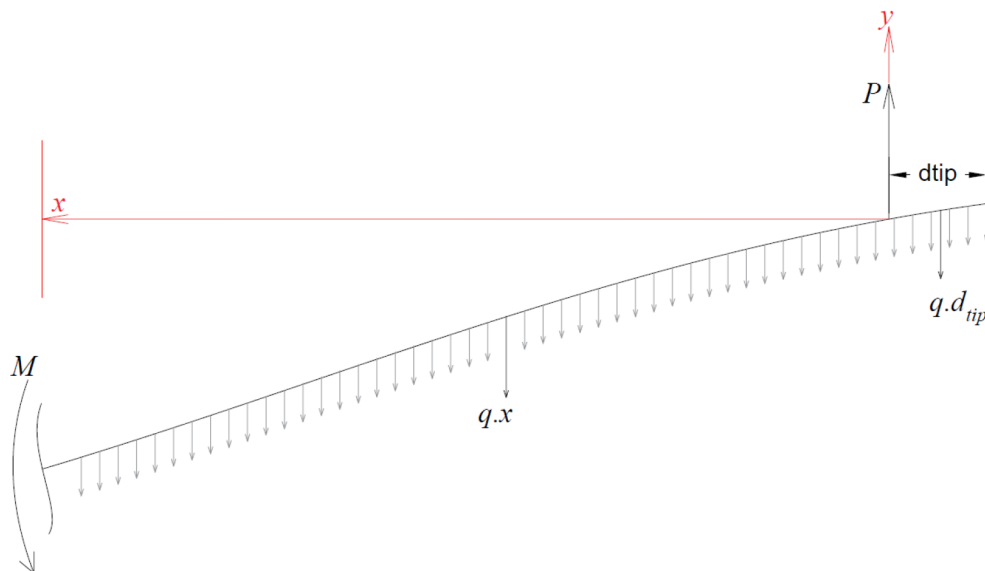


Figure 3—Free-body diagram for summing moments around an arbitrary point x

To achieve equilibrium, we sum the moments at the section point, located a distance (x) from the lifting force, and set the resulting equation to zero. This yields the following equation:

$$M - qx \cdot \frac{x}{2} + Px - qd_{tip} \left(\frac{d_{tip}}{2} + x \right) = 0 \quad (1)$$

By applying boundary conditions at TDP, where the bending moment (M) equals zero at $x = L_f$, and substituting these conditions into equation (1), we derive the following quadratic equation with L_f as the unknown variable:

$$\frac{q}{2} L_f^2 - (P - qd_{tip}) \cdot L_f + q \frac{d_{tip}^2}{2} = 0 \quad (2)$$

which corresponds to the typical quadratic equation:

$$ax^2 + bx + c = 0 \quad (3)$$

Therefore, to solve this equation for L_f , we use the quadratic formula:

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad (4)$$

Comparing equations 2 and 3:

$$\begin{aligned} x &= L_f \\ a &= \frac{q}{2} \\ b &= -(P - q \cdot d_{tip}) \\ c &= q \frac{d_{tip}^2}{2} \end{aligned}$$

then, substituting in the quadratic formula, we obtain L_f as follows:

$$L_f = \frac{(P - qd_{tip}) + \sqrt{(P - qd_{tip})^2 - q^2 d_{tip}^2}}{q} \quad (5)$$

all the parameters in the right-hand side (RHS) of Equation 5 are known.

Finally,

$$L = L_f + d_{tip} \quad (6)$$

where d_{tip} is an input parameter determined and controlled by engineers, to reach an acceptable tip angle, facilitating the subsea construction operation.

With these formulas (5 and 6), we have obtained the touchdown point distance, which we shall use in subsequent analysis.

Energy Methods

Energy methods are a powerful tool for deriving exact and approximate solutions to various structural problems. In this study, we will employ the Minimum Potential Energy method, assuming the deflection angle is sufficiently small to treat the problem as linear.

Consequently, the nonlinearity, whether moderate or full, is entirely disregarded. As will be demonstrated, the results derived will be highly satisfactory compared to those obtained by commercial software.

In the context of the bending theory of beams, the total potential energy is

$$\Pi = \int_0^L \frac{EI}{2} (\omega'')^2 dx - \int_0^L q(x) \omega dx \quad (7)$$

where

ω'' represents the second derivative of the vertical displacement or the curvature,

E is Young's modulus,

I is the (composite) moment of inertia, and

$q(x)$ is the function of the distributed load, which, in our case, is assumed constant (q) and refers to the weight per meter.

To determine the pipeline profile, we assume an initial solution with an appropriate number of unknown variables, then proceed with the analysis to obtain these unknowns. The assumed solution should be judiciously selected from various options, including, for instance:

- i. Polynomial representation or Taylor series expansion
- ii. Fourier series expansion

We will assume the solution as a Taylor series expansion and then apply the stationary condition to derive the equation coefficients.

The stationary condition is expressed as:

$$\frac{\partial \Pi}{\partial a_n} = 0 \quad (8)$$

where a_n represents a_0, a_1, a_2 , etc. (see equation 10.)

Pipeline Profile

We assume the pipeline profile to be a power series as follows:

$$\omega(x) = \sum_{n=1}^N a_n x^n \quad (9)$$

This is Taylor's expansion, which is the best fit for problems involving a cantilever beam under a point force at the tip.

We will consider only five terms as follows:

$$\omega(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 \quad (10)$$

where $\omega(x)$ represents the vertical displacement.

As in a typical cantilever problem, the displacement and deflection angle vanish at the fixed support, or in our case, at the TDP. This is expressed as:

$$\begin{aligned}\omega(x=0) &= 0, \text{ and} \\ \omega'(x=0) &= 0\end{aligned}$$

Substituting these conditions into Equation 10, we find that a_0 equals zero. Furthermore, differentiating Equation 10 to obtain the slope equation $\omega'(x)$ and applying the boundary conditions, we find that a_1 also equals zero.

Thus, Equation 10 can be reduced to:

$$\omega(x) = a_2x^2 + a_3x^3 + a_4x^4 \quad (11)$$

The first and second derivatives with respect to x can be expressed as:

$$\omega'(x) = 2a_2x + 3a_3x^2 + 4a_4x^3 \quad (12)$$

$$\omega''(x) = 2a_2 + 6a_3x + 12a_4x^2 \quad (13)$$

where $\omega''(x)$ represents the curvature.

Substituting Equations 11 and 13 into Equation 7 (total potential energy equation), we obtain:

$$\Pi = \frac{EI}{2} \int_0^L (2a_2 + 6a_3x + 12a_4x^2)^2 dx - q \int_0^L (a_2x^2 + a_3x^3 + a_4x^4) dx - P(a_2L_f^2 + a_3L_f^3 + a_4L_f^4) \quad (14)$$

As q is assumed constant, it can be taken out of the integral. Integrating, rearranging, and reducing, we reach the expression of Π in terms of the unknown coefficients a_2 , a_3 , and a_4 .

Differentiating Π partially with respect to a_2 , a_3 , and a_4 and equating to zero, we obtain three equations with three unknowns. This can be expressed in matrix form as:

$$\begin{bmatrix} 4L & 6L^2 & 8L^3 \\ 6L^2 & 12L^3 & 18L^4 \\ 8L^3 & 18L^4 & 28.8L^5 \end{bmatrix} \begin{bmatrix} a_2 \\ a_3 \\ a_4 \end{bmatrix} = \frac{1}{EI} \begin{bmatrix} PL_f^2 - q\frac{L^3}{3} \\ PL_f^3 - q\frac{L^4}{4} \\ PL_f^4 - q\frac{L^5}{5} \end{bmatrix} \quad (15)$$

Using matrix algebra, we solve for a_2 , a_3 , and a_4 , thereby determining the vertical displacement field, or in other words, the pipeline profile in static equilibrium.

Stress analysis

Upon evaluating the pipeline profile, the tip height and the tip angle are calculated by substituting L in equations 11 and 12, respectively. These two parameters may or may not fulfill the requirements for lifting from a construction perspective. Therefore, if the desired lifting conditions are not met, adjustments can be made by either increasing or decreasing the lifting force or altering its location (d_{tip}).

We can also obtain the bending stress and compare it with the allowable stress. In some cases, before making such a comparison, we may calculate the combined stress, considering both the tensile and hoop stresses. However, in a slight lift like this, tensile stress can usually be neglected, along with hoop stress, especially in shallow waters.

To find the maximum bending stress, we follow the following steps:

1. Write the equation of the bending moment as follows:

$$M = EI\omega''(x) = EI[2a_2 + 6a_3x + 12a_4x^2] \quad (16)$$

2. To find the x value where M is maximum, we get the first derivative of M and equate it to zero as follows:

$$M' = EI[6a_3 + 24a_4x] = 0 \quad (17)$$

3. Solving for x in 17, we obtain:

$$x(at M_{max}) = -\frac{a_3}{4a_4} \quad (18)$$

4. Compute the maximum bending moment by substituting 18 into 16

5. The maximum bending stress can now be calculated by mechanics of materials (Hibbeler, 2011) as follows:

$$\sigma_b = \frac{M_{max}r}{I_{steel}} \quad (19)$$

where

r is the steel pipe radius, and

I_{steel} is the moment of inertia of the steel cross-section.

Note the difference between (I) and (I_{steel}). The moment of inertia (I), used previously throughout the calculation, must be the composite moment of inertia, considering the concrete coating, if any.

6. If necessary, the tensile stress can be calculated by multiplying the weight per meter (q) by the height of the pipe at x (M_{max}) calculated in step 3 above, and then dividing by the cross-sectional area of the steel. Remember to subtract the compression induced by the water column on the pipeline's free end. Nevertheless, as mentioned earlier, the tensile stress is too small compared to the bending stress due to the pipeline's small height.
7. Additionally, the hoop stress can be calculated, but it can also be neglected when compared to the bending stress. If both tensile and hoop stress are calculated, the combined stress should be evaluated by the Von Mises criterion.

Computer Application

One of the powerful tools in the domains of mathematics, science, and engineering is the program *Mathematica*, also known as *Wolfram Mathematica*. It is based on the Wolfram programming language, which empowers engineers and scientists to generate computer code that can be developed into easy-to-use applications operating within the environment of Mathematica. It can also be further developed into standalone web applications, which can be shared and utilized by others through web links.

The above derivation has been translated into a Wolfram computer code and application – see the snapshot below.

With this application, users can input data on the left side and immediately obtain results on the right side. Furthermore, the new result is reflected instantly on the graph, providing both written and visual outcomes in real-time.

The following figure shows a snapshot of the web application, which performs the same tasks but takes one or two extra seconds, depending on the internet speed.

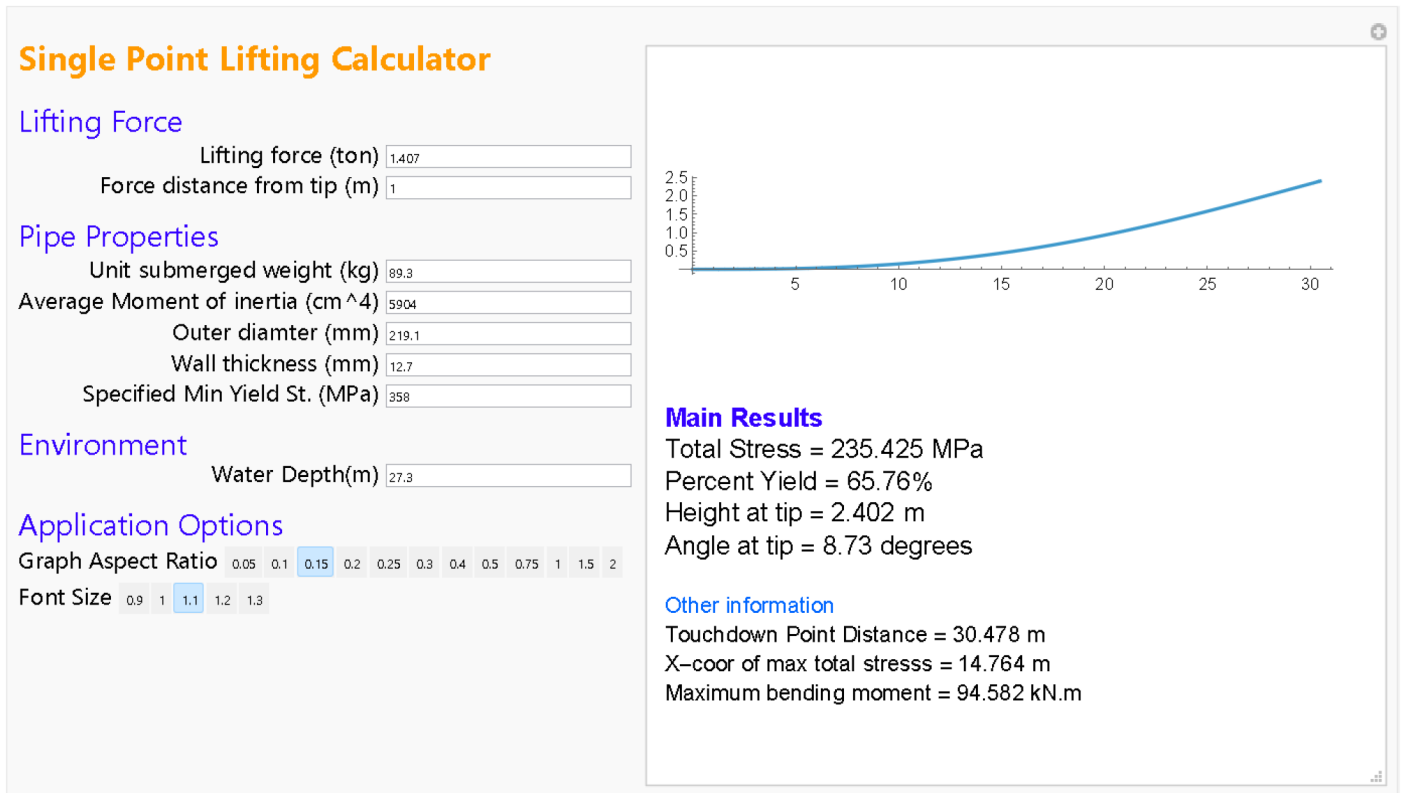


Figure 4—Snapshot for a single-point lifting application developed by Wolfram Language

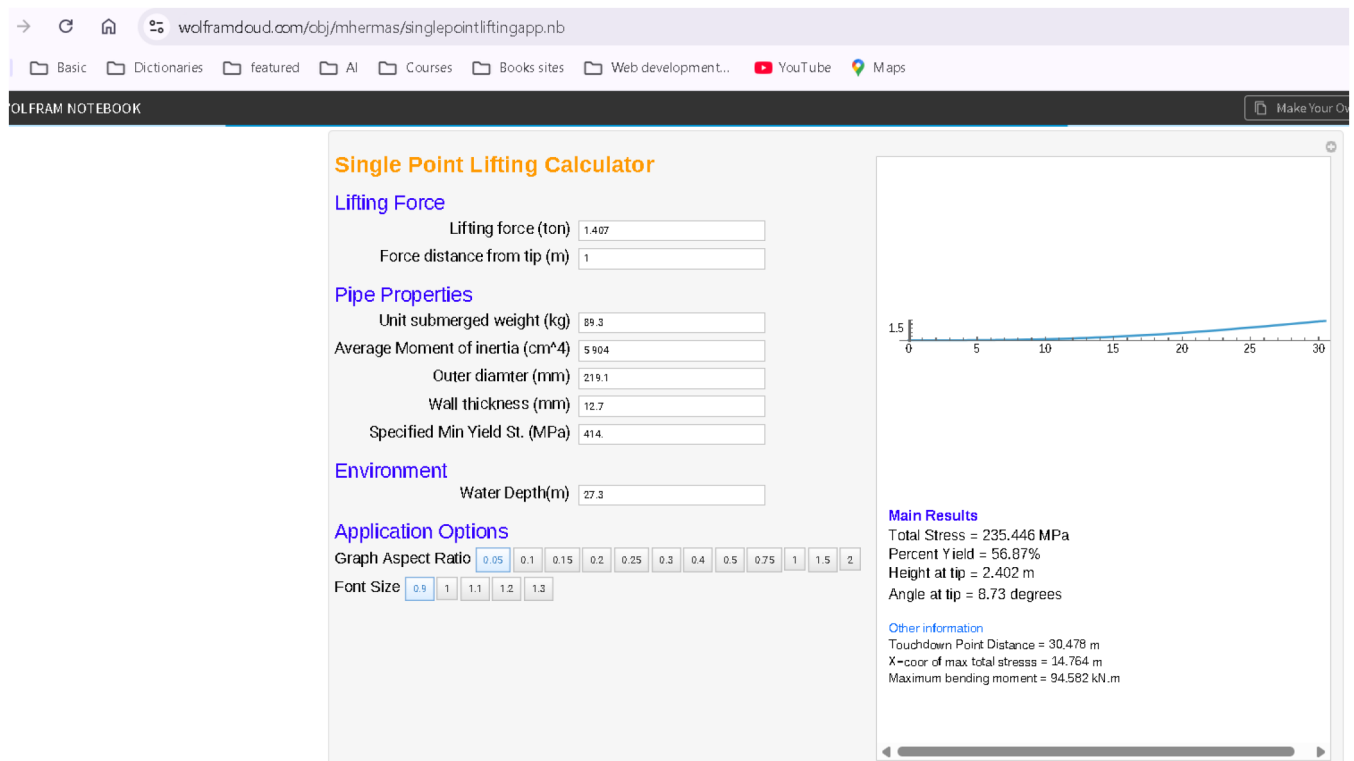


Figure 5—Web application

Commercial software and results comparison

Real project study by OFFPIPE

A case study was conducted using real data from a previous project, where the single-point lift calculation was performed using OFFPIPE, a globally recognized software. OFFPIPE is based on FEM and provides exact solutions (theoretically approximate) mainly for pipelay stress analysis, so its results are accepted as a benchmark, whereby we can compare them with other prototype software. A brief introduction to the finite element method is provided in the following subsection.

The input data for the case study can be summarized as follows:

- Coated pipe average moment of inertia: **5903.90 cm⁴**
- Weight per-unit-length submerged: **875.60 N/m**
- Steel outside diameter: **219.1 mm**
- Steel wall thickness: **12.7 mm**
- Yield stress: **358 MPa**
- Water depth: **27.30 m**

The comparison between the OFFPIPE and the prototype application results can be tabulated as follows:

Table 1—Comparison between OFFPIPE and Prototype app results

Criteria	OFFPIPE	Wolfram app	Error
Total stress	233.79	235.43	0.7%
Percent Yield	65.31	65.76	0.7%
Height at tip	2.4	2.4	0.0%
Angle at tip	8.83	8.73	1.1%
TDP distance	30	30.48	1.6%
Maximum Bending Moment	93.91	94.58	0.7%

As shown in the preceding table, the error is minimal. This is despite the adoption of the infinitesimal deflection beam theory, which inherently disregards nonlinearity in its calculations. The accuracy of these results can be attributed to the small angle at the beam's tip, which is only 8.83 degrees.

Finite element method (FEM)

The advances in mathematics in the 1950s were closely tied to those in computer science. One of the most impressive inventions in mathematics is the numerical methods, particularly the finite element method. This method is not only a last resort for solving some differential equations but also provides an easy way to write computer code, thanks to its systematic approach, which leaves no room for logical thinking, making it a good fit for software development.

The basic idea in the FEM is to find the solution to a complicated problem by replacing it with a simpler one (Rao, 2018). Therefore, we will only be able to find an approximate solution, rather than the exact solution. The existing mathematical tools are insufficient to find the exact solution for most practical problems. Thus, in the absence of any other convenient method, we must use the FEM.

In the finite element method, the solution region is built of many small, interconnected subregions (elements). As an example of how an FE model might be used to represent a complex geometrical shape,

consider the aircraft body. Since it is very difficult to find the exact response of the plane under any specified loading condition, this structure is approximated as composed of several pieces, whereby a convenient approximate solution is assumed, and the conditions of overall equilibrium of the structure are derived. Satisfying these conditions will yield an approximate solution for the displacements and stresses.

Conclusions and Recommendations

The progression of programming and software technologies has naturally culminated in the development of sophisticated programs. Within the offshore industry, numerous software applications deliver precise solutions based on numerical methods, predominantly the Finite Element Method (FEM). On the other hand, some engineering tasks often necessitate only minimal manual verification, a spreadsheet, or a piece of computer code, rendering the immediate application of software superfluous.

However, in many scenarios, the calculations are too simple to warrant the use of extensive commercial software, yet too complex to be executed using MS Excel or Mathcad. In such cases, commercial software is often employed in reality, which can require substantial time and resources. This study underscores the significance of developing small-scale applications that are ideally suited for situations where pipeline displacement is limited, i.e., where there is no significant geometrical nonlinearity. Developing these applications in-house is advantageous as it obviates the need for specialized training, unlike commercial software. Consequently, field and project engineers can obtain results promptly and without the complexities associated with software installation or the necessity for a specialist.

Mathematically speaking, the finite element method is not always necessary; sometimes, simple analytical methods are sufficient, and more importantly, are suitable for developing small programs. Accordingly, within the offshore industry, it is imperative to adopt concrete measures towards computer-aided engineering (CAE) that transcends conventional spreadsheets as well as the traditional software packages.

Engineers must harness the potential of computing platforms, such as MATLAB, Mathematica (Wolfram language), and Python, to develop these small applications. By doing so, the working environment will become more dynamic and flexible, enabling site engineers to utilize in-house "small" programs that are user-friendly and capable of performing tasks typically managed by large commercial software.

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